GGG–T–669c
May 1, 1964

SUPERSEDING
Int. Fed. Spec. GGG-T-00669b (GSA-FSS)
August 29, 1961
Fed. Spec. GGG-T-669a
May 5, 1954

FEDERAL SPECIFICATION

TRIMMER, TREE, POLE-MOUNTED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers long-reach hand-operated tree trimmers capable of cutting branches of 1, 1-1/4, and 1-1/2 inches in diameter. It also covers pole-mounted saw blades for cutting large branches.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and styles.* Tree trimmers shall be of the following types and styles, as specified (see 6.2):

Type I.— Trimmer, rope and pulley operated.
Style A.—Center cut.
Style B.—Side cut.

Type II.—Trimmer, rope and level operated.
Style A.—Center cut.
Style B.—Side cut.

Type III.—Combination trimmer and saw.
Style A.—Center cut.
Style B.—Side cut.

1.2.2 *Sizes.* The trimmer and blade sizes shall be as shown in tables I and II, as specified.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

T-R-571—Rope, Cotton.

T-R-605—Rope, Manila and Sisal.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas

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City, Mo., Dallas, Denver, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-W-6110—Wood, Determination of Moisture Content of.

MIL-H-15424—Hand Tools, Packaging of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Lumber.* All lumber shall be of a moisture content not exceeding 15 percent at time of fabrication, shall have a slope of grain not exceeding 1 inch in 15 inches, and shall be free from defects except that slight stains, streaks, and discolorations will be allowed.

3.1.1.1 *Species.* Unless otherwise specified (see 6.2), the species of wood used for pole-mounted trimmers shall be of basswood, Sitka spruce, or Douglas fir and shall have not less than six annual rings per inch. The sections of the handle for any one trimmer shall be of the same species.

3.1.2 *Springs.* All coil springs shall be of high quality steel such as music wire, hard drawn spring steel wire, or oil tempered steel wire.

3.1.3 *Rope.* All rope shall be not less than 3/8-inch nominal diameter, and shall be either grade A manila in accordance with T-R-605 or cotton in accordance with T-R-571. A rope shall be furnished with each pole-mounted trimmer. The rope shall be of sufficient length in order that the trimmer may be efficiently operated. In lieu of rope, braided cotton sash cord of No. 8 minimum size may be used.

3.1.4 *Cutting blade.* The cutting blade shall be of high-grade carbon tool steel and shall show a hardness of not less than 48 nor more than 56 on the Rockwell C scale, the reading to be taken as near the cutting edge as practicable. The cutting blade shall have a ground and polished keen cutting edge.

3.1.4.1 *Cutting-blade bolt.* The bolt shall be of high-grade steel properly heat-treated to withstand wear and deformation under maximum usage. Adequate means shall be provided for preventing loss of proper tightness and also for easy removal in order to resharpen the cutting blade.

3.1.5 *Saw blades.* All saw blades shall be of high-grade commercial saw steel and shall show a hardness of not less than 44 nor more than 52 on the Rockwell C scale. The reading shall be taken as near to the cutting edge as practicable. Each blade shall be curved, properly sharpened for fast cutting on the draw stroke, and designed to prevent binding. Except on the saw-blade attachment for type II trimmers, all blades shall have a bolt hole and three locating holes to accommodate the various angles of adjustment. The blades shall be free from any defects that may adversely affect serviceability.

3.2 Construction of pole-handle sections. Each complete pole mounted trimmer shall consist of either two or three pole handle sections. The maximum length of any single

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section shall not exceed nine feet. The sections shall be provided with positive locking ferrules of the telescoping type, except that type II may have a single ferrule or when sections are furnished they may be put together by a two piece channel type connector secured by bolts and nuts, and be so constructed as to prevent the head or any section from working loose or coming apart in service.

3.2.1 *Finish of wood handles and sections.* All handles and sections shall be smoothly finished, and corners and edges rounded so as to be comfortable to the hands. They shall be adequately covered with two durable coats of transparent varnish or shellac to prevent absorption of moisture and permit visible inspection of the wood.

3.3 General design and construction. The design of the trimmers shall be such that only normal manual effort is required by the operator to completely sever the maximum diameter branches specified. The trimmers shall be so constructed that no part will become detached while in use in normal service.

3.4 Identification marking. All trimmers shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. When specified (see 6.2), the trimmers shall be marked in accordance with MIL-STD-130.

3.5 Type I, rope and pulley operated.

3.5.1 *Style A, center cut.* Style A trimmers shall be of the center-cut action and shall be hand operated by means of a rope and chain or one-piece rope over a pulley which is mounted on the outer end of the lever-actuated cutting blade. The trimmer shall consist essentially of a cutter head and pole-handle sections.

3.5.1.1 *Cutter head.* The head shall be malleable iron, welded steel stampings, or drop-forged steel, having a curved hook at one end to cradle the branch and a socket or fork at the other end into which the top pole-handle section is inserted and securely fastened. Adequate means shall be provided for mounting the lever-actuated cutting blade in such a manner as to allow the cutting edge to pass through the hook opening when the lever is drawn down. Sufficient spring tension shall be provided to secure the cutting blade in an open position when the rope tension is released after the cutting operation.

3.5.1.1.1 *Finish.* Heads designed for center cut action shall be smooth and free of scale. The heads shall be protected against rust in accordance with the manufacturer's standard practice.

3.5.1.2 Type I, style A tree trimmers shall conform to table I as specified (see 6.2), and shall be similar to figure 1.

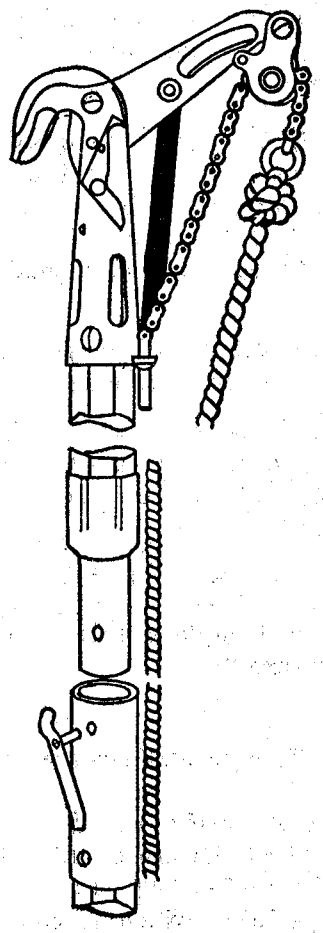


FIGURE 1.—Type I, style A, rope and pully operated, center-cut tree trimmer.

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TABLE I.—*Type I, style A, rope and pully operated, center-cut tree trimmers*

Size, or branch (diameter) capacity	Nominal over-all length	Cross-sectional area of pole (min.)	Weight (max.)
Inches	*Feet*	*Square inches*	*Pounds*
1	18	1.25	9
1-1/2	18	1.85	18

3.5.2 *Style B, side cut.* Style B trimmers shall conform to 3.5.1 except they shall be of the side-cut action and shall sever branches up to 1-1/4 inches in diameter.

3.5.2.1 *Cutter head.* The cutter head shall conform to 3.5.1.1 except that the cutting blade shall be mounted in such a manner that will allow the cutting edge to pass along the side of the hook when the lever is drawn down.

3.5.2.1.1 *Finish.* Heads designed for side cut action shall be hardened and ground flat over the area that the cutter blade operates, and shall have a smooth, natural surface, free of scale over the remaining surface.

3.5.2.2 *Nominal overall length.* The over-all length shall be approximately 16 feet.

3.5.2.3 *Cross-sectional area of pole.* The minimum cross-sectional area of the wood pole shall be 1.25 square inches.

3.5.2.4 *Weight.* The maximum weight of style B trimmers shall be 10 pounds.

3.5.2.5 Type I, style B trimmers shall be similar to figure 2.

3.6 Type II, rope and lever operated.

3.6.1 *Style A, center cut.* Style A trimmers shall be of the center-cut action and shall be hand operated by means of a rope attached to a lever which is connected to the cutting blade or cutting blade lever with a metal rod. The trimmer shall consist essentially of a cutter head and pole handle sections and shall sever branches up to a diameter of 1 inch.

3.6.1.1 *Cutter head.* The head shall be either drop-forged steel or steel stampings which are riveted or welded. The head shall have a curved hook at one end to cradle the branch and a fork at the other end into which the top pole handle section is inserted and securely fastened. Adequate means shall be provided for mounting the lever-actuated cutting blade in such a manner as to allow the cutting edge to pass through the hook opening when the lever is drawn down. Sufficient spring tension shall be provided to secure the cutting blade in an open position when the rope tension is released aft-

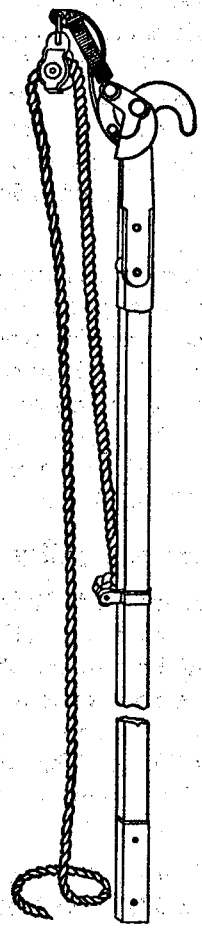


FIGURE 2.—Type I, style B, rope and pully operated, side-cut tree trimmers.

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er the cutting operation. Cutting blades and bolts shall conform to 3.1.4 and 3.1.4.1, respectively.

3.6.1.1.1 *Finish.* The finish shall conform to 3.5.1.1.1.

3.6.1.2 *Pole-handle sections.* The top section shall provide means for securely mounting a plain or handle-type lever which has a metal rod connector for operating the cutting blade. The plain or handle-type lever shall have a suitable hole in the outer end for attaching the rope. The sections shall otherwise conform to 3.2 and 3.2.1.

3.6.1.3 *Nominal overall length.* The overall length shall be approximately 12 feet.

3.6.1.4 *Cross-sectional area of pole.* The minimum cross-sectional area of the wood pole shall be 0.90 square inch.

3.6.1.5 *Weight.* The maximum weight of style A trimmers shall be 7 pounds.

3.6.1.6 *Saw-blade attachment.* When specified (see 6.2), a saw blade conforming to 3.1.5 except that it has no angle adjustment and can be easily attached to the cutter head, shall be furnished. The saw blade attachment on the cutter head shall not cause interference with the proper cutting blade action. The length of the saw blade shall be approximately 12 inches.

3.6.1.7 Type II, style A trimmers shall be similar to figure 3.

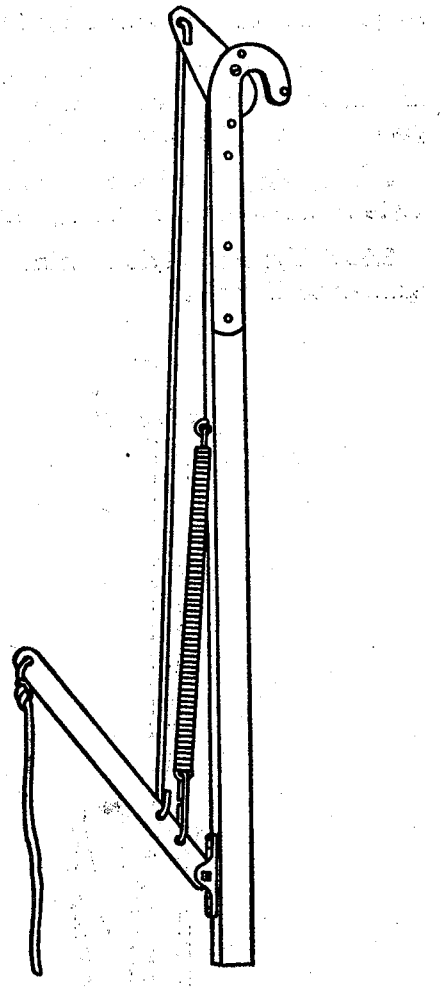


FIGURE 3.—Type II, style A, rope and lever operated, center-cut tree trimmers.

3.6.2 *Style B, side cut.* Style B trimmers shall conform to 3.6.1 except they shall be of the side-cut action.

3.6.2.1 *Cutter head.* The head shall be either drop-forged steel or steel stampings which are riveted or welded. The head shall have a curved hook at one end to cradle the branch and a fork at the other end into which the top pole-handle section is inserted and securely fastened. Adequate means shall be provided for mounting the lever-actuated cutting blade in such a manner as to allow the cutting edge to pass along the side of the hook when the lever is drawn down. Sufficient spring tension shall be provided to secure the cutting blade in an open position when the rope tension is released after the cutting operation.

3.6.2.1.1 *Finish.* The finish shall conform to 3.5.1.1.1.

3.6.2.2 *Pole-handle sections.* The top section shall provide means for securely mounting a plain or handle-type lever which has a metal rod connector for operating the cutting blade. The plain or handle-type lever shall have a suitable hole in the outer end for attaching the rope. The sections shall otherwise conform to 3.2 and 3.2.1.

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3.6.2.3 *Nominal overall length.* The overall length shall be approximately 12 feet.

3.6.2.4 *Cross-sectional area of pole.* The minimum cross-sectional area of the wood pole shall be 0.90 square inch.

3.6.2.5 *Weight.* The maximum weight of style B trimmers shall be 8 pounds.

3.6.2.6 Type II, style B trimmers shall be similar to figure 4.

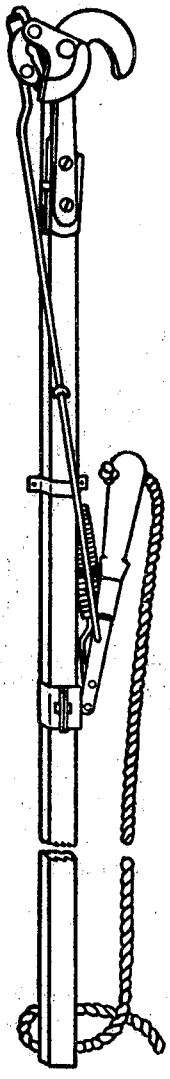


FIGURE 4.—Type II, style B, rope and lever operated, side-cut tree trimmer.

3.7 Type III, combination trimmer and saw.

3.7.1 *Style A, center cut.* Style A trimmers shall conform to 3.5.1, except a saw head and saw blade shall be furnished in addition to the cutter head and pole-handle sections. The size or branch diameter capacity, nominal overall length, cross-sectional area of pole, and weight of the trimmer shall conform to table I, as specified, and the trimmer shall be similar to figure 1.

3.7.1.1 *Saw head.* The saw-head section shall be forged steel, steel stampings, or aluminum alloy, having a curved hook at one end for pulling out loose branches and a socket or fork at the other end into which the top pole-handle section is inserted and securely fastened. The head shall be provided with a positive locating pin in order to adjust the saw blade to various angles, and shall also have a bolt and wing nut for securing the saw blade.

3.7.1.2 *Saw blade.* The saw blades shall conform to 3.1.5 and table II, as specified.

TABLE II.—*Saw blades for type III, styles A and B, combination trimmer and saw*

Nominal length	Number of teeth per inch	Type of teeth
Inches		
14	8	Needle point, peg
15	6-1/2	Needle point, peg
15	8	Needle point, peg
16	8	Needle point, peg
16	7	Needle point, peg
18	6	Needle point, peg

3.7.1.3 The saw head and saw blade of type III, style A combination trimmer and saw shall be similar to figure 5.

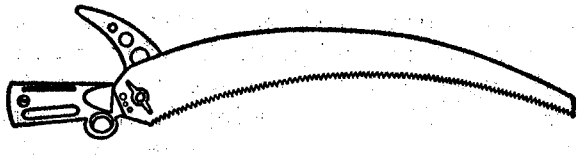


FIGURE 5.—Saw head and blade for type III, style A, combination trimmer and saw, center cut.

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3.7.2 *Style B, side cut.* Style B trimmers shall conform to 3.5.1, except they shall be of the side-cut action, sever branches up to 1-1/4 inches in diameter, and be furnished with a saw head and saw blade in addition to the cutter head and pole-handle sections. The cutter head and finish, pole-handle sections, nominal overall length, cross-section-type III, style A combination trimmer and al area of pole, and weight shall conform to 3.5.2.1, 3.5.2.1.1, 3.5.2.2, 3.5.2.3, 3.5.2.4, and 3.5.2.5, respectively.

3.7.2.1 *Saw head.* The saw head shall conform to 3.7.1.1.

3.7.2.2 *Saw blades.* The saw blades shall conform to 3.1.5 and table II, as specified.

3.7.2.3 Type III, style B trimmers shall be similar to figure 2, and the saw head and saw blade shall be similar to figure 6.

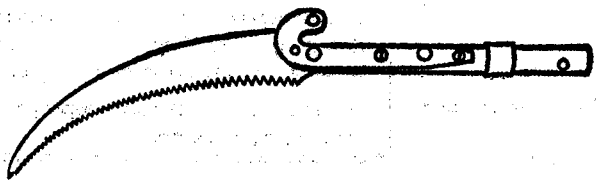


FIGURE 6.—Saw head and blade for type III, style B combination trimmer and saw, side cut.

3.8 **Repair parts.** The contractor shall furnish repair parts when specified (see 6.2.1). All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, and replaced or adjusted without requiring modification.

3.9 **Workmanship.** Workmanship shall be first class in all respects. Trimmers shall be smoothly and properly finished, and free from any defects which may adversely affect their durability and serviceability.

3.10 **Illustrations.** The illustrations herein are descriptive and not restrictive, and are not intended to preclude the purchase of trimmers otherwise conforming to this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for lot acceptance.**

4.2.1 *Lot formation.* For purposes of acceptance inspection, a lot shall consist of all trimmers of one type and style presented at one time.

4.2.2 *Sampling for examination.* Sampling for visual and dimensional examinations shall be performed in accordance with the requirements of MIL-STD-105. The inspection level and acceptable quality level shall be level II and 2.5 respectively.

4.2.3 *Sampling for tests.* A random sample of tree trimmers shall be taken from each inspection lot in accordance with the requirements of MIL-STD-105. The inspection level and acceptable quality level shall be S-4 and 1.5 respectively.

4.3 **Inspection.**

4.3.1 *Visual and dimensional examinations.* Each of the sample trimmers selected in accordance with 4.2.2 shall be visually examined in accordance with the following classification of defects.

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		Classification of defects
101	Handles	Finish not as specified. Slope of grain more than 1-inch in 15-inches; not specified species. Dimensions not as specified.
102	Cutter head	Material not as specified. Components missing or malformed.
103	Workmanship	Assembly incorrect or malfunctioning.
104	Weight	Not within specified limits.

4.3.2 *Testing.*

4.3.2.1 *Moisture content.* A random sample of handles selected in accordance with 4.2.3 shall be tested for moisture content. This test will be conducted prior to application of the coating and shall be conducted in accordance with the requirements of MIL-W-6110.

4.3.2.2 *Cutting test.* Sample trimmers selected in accordance with 4.2.3 shall be tested for cutting ability by severing at least 15 branches of not less than the specified capacity (branch diameter) of green wood such as ash, beech, hickory, or pine. After being subjected to this test, the trimmers and shears shall show no sign of distortion to any part, damage to the cutting edges, or indication of failure in any respect. The trimmers shall be well balanced, operate smoothly, and sever branches with a minimum of effort. Trimmers having a saw blade shall make at least 5 cuts through branches of not less than 2 inches in diameter of the aforementioned green wood. The saw blade shall show no sign of distortion, damage to the teeth or blade or significant binding during the sawing operation. Sample trimmers failing any of these tests shall be rejected.

4.3.2.3 *Hardness test.* Cutting blades and saw blades of trimmers selected for the cutting test shall be tested for compliance with the hardness requirements specified in 3.1.4 and 3.1.5 respectively. Any cutting blade or saw blade which does not meet the respective hardness requirements shall be rejected.

4.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of the applicable specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-1 and the acceptable quality level shall be 4.0 defects per 100 units.

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling; bulging or distortion of containers.
Preservation	Preservation improper, improperly applied, or missing.

5. PREPARATION FOR DELIVERY

(Civil agencies. Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Level A preservation and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* The trimmers shall be preserved and packaged in a manner to afford adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 *Level A.* Level A packing shall be in accordance with level A of MIL-H-15424.

5.2.2 *Level B.* Level B packing shall be in accordance with level B of MIL-H-15424.

5.2.3 *Level C.* Trimmers, cleaned, preserved and packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or order (see 6.2), shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking specified in the contract or order (see 6.2), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Pole-mounted trimmers are intended to be used for cutting tree branches from 18 to 24 feet above the ground. By use of the hook-head, the branches can be pulled down when they cling to uncut branches. Saw blades for trimmers are intended to sever branches too large for the capacity of the cutting blade.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and style (when applicable) required (see 1.2.1).

(*c*) Species of wood, if different (see 3.1.1.1).

(*d*) Size of trimmer when type I or III, style A is required (see table I and 3.7.1).

(*e*) Saw blade attachment when required for type II, style A (see 3.6.1.6).

(*f*) Nominal length of blade and number of teeth per inch, when type III trimmers are required (see table II).

(*g*) Whether preparation for immediate use, domestic storage, or overseas shipment is required (see 5.1).

6.2.1 When ordering repair parts, the following should be specified:

(*a*) The manufacturer's name or trademark of the trimmer for which the parts are required.

(*b*) Type, style, and size of trimmer for which the part is required.

(*c*) When saw blades are required, specify length of blade and number of teeth.

(*d*) Name and description, if necessary, of the part required.

(*e*) Quantity of each item required.

6.3 Superseding data. Type IV pruning shears formerly in this specification (GGG-T-669) are covered by Federal Specification GGG-S-285.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Pruners, tree.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

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Motor:

Pruners, tree.

Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification

MILITARY CUSTODIANS:

Air Force—MOAMA

Army—MO(ERDL)

Navy—YD

DCSC

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